

Design Evaluation Report – IP3: Virtual Retail Shopping

Objective and Validation Metrics

This phase evaluated the usability, immersion, and intuitiveness of the XR retail prototype, enabling users to grab, snap, and confirm clothing selections through hand-tracking and gestures. Building on IP2's colour-changing prototype, this version introduced hand-based grabbing, snap-to-mannequin interaction, a thumbs-up confirmation gesture, and an on-screen text notification. The study examined action smoothness, snapping accuracy, gesture detection, and feedback clarity.

Participant Demographics

Four participants aged 25–44 (3 males, 1 female) tested the prototype, ranging from beginner to advanced XR familiarity.

Interaction Evaluation Findings

All users found the grabbing natural and intuitive, showing improvement from controller-based inputs. However, snapping was inaccurate and unrealistic, reducing immersion. The thumbs-up gesture worked but appeared small and below eye level. The “Added to Cart” text appeared successfully, but facing in just one direction. Two users found the overall grab–snap–gesture flow natural, yet all felt confident and enjoyed the experience, confirming that gesture-based interaction is engaging and intuitive.

Open-Ended Feedback

Cube-grabbing and colouring were the most satisfying interactions due to instant visual feedback. No difficulties were reported with grabbing or gestures. Users suggested rotating the feedback text to face the user and improving the layout, as some elements appeared distant. Enhancing textures and lighting was also recommended for realism.

Key Insights

1. Hand-tracking enhanced control and engagement.
2. Snapping realism required improvement.
3. Feedback placement affected clarity.
4. Spatial layout influenced usability and immersion.

Evaluation Summary

Goal	Outcome	Summary
Implement hand-based interaction	Achieved	Intuitive and comfortable.
Improve realism & immersion	Partial	Snapping misaligned; distant environment.
Provide clear feedback	Partial	Text visible but poorly positioned.
Maintain usability & confidence	Achieved	Users operated independently.
Ensure engagement & enjoyment	Achieved	All preferred gesture-based XR shopping.

Reflection

This iteration advanced from IP2's static design to gesture-driven interaction. Despite minor issues, users' confidence validated the approach. Gesture-based control made the experience more personal and realistic, marking progress toward immersive XR retail interaction.

Final Reflection:

Prototype Session Review (IP3)

The IP3 test achieved natural, gesture-driven flow. Grabbing worked well, confirming the benefits of hand tracking, while snapping misalignment reduced realism. The thumbs-up gesture and feedback text confirmed actions but required better visibility. Half found the flow fully natural, though all completed tasks independently and enjoyed the session.

Methodological Reflection

A mixed-methods approach—questionnaires and observation—captured both performance and perception. Rapid prototyping produced actionable feedback, such as gesture visibility and snapping precision. Although consistency in lighting and space could be improved, the method effectively revealed embodied user insights in XR.

Concept Evaluation

The concept was partially validated. Users valued natural gesture-based shopping over controllers, confirming the interaction model's potential. However, snapping inaccuracies and feedback placement limited realism. The concept proved effective for human-like digital shopping but needs refinement for full immersion.

Improvements and Extensions

Future iterations should refine snapping alignment, enlarge gesture visuals, and reposition feedback for visibility. Exploring AR could merge physical and virtual retail—allowing real-space garment previews. Adding haptic and voice feedback could enhance multimodal realism. This confirms gesture-driven interaction as a strong foundation for immersive XR retail.